

TWO PARALLEL QUEUES CREATED BY ARRIVALS WITH TWO DEMANDS II*

LEOPOLD FLATTO†

Abstract. We consider the double queue that arises when arriving customers simultaneously place two demands handled independently by two servers. It is assumed that the customer arrivals form a Poisson process with mean 1, the servers have exponential service times with rate α, β , and $1 < \alpha \leq \beta$, which insures stability of the queue. Let X_1, X_2 be the respective lengths of the α - and β -queues, and $p_{mn} = P[X_1 = m, X_2 = n]$ at equilibrium. In a previous paper with the same title we obtained a formula for the generating function $P(z, w) = \sum p_{mn} z^m w^n$. We use this to derive the asymptotic behavior of p_{mn} as $m, n \rightarrow \infty$. The asymptotic results are employed to study the interdependence of X_1, X_2 . We derive limit laws for the expectation and distribution for either of these random variables conditioned on the other.

1. Introduction and summary. We continue the study of the double queue considered in [2]. We use the notation and terminology of that paper and assume its results. The double queue under consideration is the one created by arriving customers who simultaneously place two demands handled independently by two servers. The arrivals form a Poisson process with mean 1 and the service times are exponential with rates α, β . It is assumed that $1 < \alpha \leq \beta$, which is equivalent to stability of the double queue. Let X_1, X_2 be the respective lengths of the queues treated by the servers with rates α, β and let $p_{mn} = P[X_1 = m, X_2 = n]$ be the equilibrium probabilities. Since $\sum p_{mn} = 1$ and $p_{mn} > 0$ for all m, n , the generating function $P(z, w) = \sum p_{mn} z^m w^n$ converges for $|z|, |w| \leq 1$. In [2] we obtained a formula for $P(z, w)$.

It appears difficult to obtain from this formula a simple revealing expression for p_{mn} . In this paper, we derive from it the asymptotic behavior of p_{mn} as $m, n \rightarrow \infty$. The asymptotic formulas are used to study the interdependence of X_1, X_2 at equilibrium. We derive the following limit laws for the expectation and distribution of either of these random variables conditioned on the other.

THEOREM 1. i) If $\alpha < \beta$, then

$$(1.1) \quad E(X_2 | X_1 = m) \sim \frac{1}{\beta/\alpha - 1} \quad \text{as } m \rightarrow \infty.$$

ii) If $\alpha < \beta$, then

$$(1.2) \quad E(X_1 | X_2 = n) \sim \left(\frac{\beta - \alpha}{\beta - 1} \right) n \quad \text{as } n \rightarrow \infty.$$

iii) If $\alpha = \beta$, then

$$(1.3) \quad E(X_2 | X_1 = m) \sim 2 \sqrt{\frac{\alpha m}{(\alpha - 1)\pi}} \quad \text{as } m \rightarrow \infty.$$

THEOREM 2. i) If $\alpha < \beta$, then

$$(1.4) \quad \lim_{m \rightarrow \infty} P[X_2 = n | X_1 = m] = \frac{\beta/\alpha - 1}{(\beta/\alpha)^{n+1}}, \quad 0 \leq n < \infty.$$

ii) If $\alpha < \beta$, then

$$(1.5) \quad \lim_{n \rightarrow \infty} P \left[X_1 < \frac{\beta - \alpha}{\beta - 1} n + x \sqrt{n} | X_2 = n \right] = \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2\sigma^2} dt, \quad -\infty < x < \infty$$

* Received by the editors March 23, 1984, and in revised form November 6, 1984.

† AT&T Bell Laboratories, Murray Hill, New Jersey 07974.

where

$$(1.6) \quad \sigma^2 = \frac{\alpha^2\beta + \alpha\beta^2 + \alpha + \alpha^2 + \beta + \beta^2 - 6\alpha\beta}{(\beta - 1)^3}.$$

iii) If $\alpha = \beta$, then

$$(1.7) \quad \lim_{m \rightarrow \infty} P[X_2 < x\sqrt{m} | X_1 = m] = \frac{1}{\sigma} \sqrt{\frac{2}{\pi}} \int_0^x e^{-t^2/2\sigma^2} dt, \quad 0 \leq x < \infty$$

where

$$(1.8) \quad \sigma^2 = \frac{2\alpha}{\alpha - 1}.$$

Theorem 1 is derived directly from the formula for $P(z, w)$, while Theorem 2 is derived from the asymptotics for p_{mn} . We remark that the quantity σ^2 of (1.6) is positive. This follows upon applying the arithmetic-geometric means inequality to the six numbers $\alpha^2\beta, \alpha\beta^2, \dots, \beta^2$ appearing in the numerator of σ^2 . The limit laws imply the somewhat surprising fact that if a given queue length is observed to be large, then the other queue length is bound to be shorter, even when served by the slower server.

To derive the asymptotics of p_{mn} , we obtain an integral formula for it and then apply the saddle point method to obtain the asymptotic behavior of the integral. This method is explained in various texts on asymptotics (see for instance [1]). It is used by Malyshev [5] in his study of random walks on positive two-dimensional lattices to obtain the asymptotics of the equilibrium probabilities. Malyshev uses topological arguments involving the notion of structural stability to prove the existence of the steepest descent curve needed to apply the saddle point method. We accomplish the same by complex variable theory.

To make the present paper self-contained, we review in § 2 the concepts and results of [2] which are needed here. The integral formula for p_{mn} is derived in § 3. In §§ 4 and 5 we describe the saddle points and the steepest descent curve associated with the saddle point method. The existence of the steepest descent curve is based upon the mapping properties of a certain function discussed in the appendix. The asymptotic formulas for p_{mn} are given in § 6. Theorems 1 and 2 are proved in §§ 7 and 8.

2. Review of [2]. Use the notation

$$\chi(A) = \begin{cases} 1 & \text{if property } A \text{ holds,} \\ 0 & \text{otherwise.} \end{cases}$$

The equilibrium probabilities p_{mn} satisfy the equations

$$(2.1) \quad \begin{aligned} [1 + \alpha\chi(m \geq 1) + \beta\chi(n \geq 1)]p_{mn} \\ = \alpha p_{m+1,n} + \beta p_{m,n+1} + \chi(m, n \geq 1)p_{m-1,n-1}, \quad m, n > 0. \end{aligned}$$

Multiply both sides of (2.1) by $z^m w^n$ and sum over m, n . We obtain the functional equation

$$(2.2) \quad Q(z, w)P(z, w) = N(z, w), \quad |z| \text{ and } |w| \leq 1$$

where

$$(2.3) \quad Q(z, w) = (1 + \alpha + \beta)zw - \alpha w - \beta z - z^2 w^2,$$

$$(2.4) \quad N(z, w) = \beta z(w - 1)P(z, 0) + \alpha w(z - 1)P(0, w).$$

The formulas for $P(z, 0)$, $P(0, w)$ are given in Theorem 2.2 and involve the roots of the z - and w -discriminants of $Q(z, w)$. We collect the following facts about these roots.

THEOREM 2.1. *Let*

$$(2.5) \quad D_1(z) = [(1 + \alpha + \beta)z - \alpha]^2 - 4\beta z^3, \quad D_2(w) = [(1 + \alpha + \beta)w - \beta]^2 - 4\alpha w^3.$$

$D_1(z)$, $D_2(w)$ are the respective z - and w -discriminants of $Q(z, w)$. The three roots a_1, a_2, a_3 of $D_1(z)$ are real and distinct with $0 < a_1 < a_2 < 1 < a_3$, $D_1(z) > 0$ on $(-\infty, a_1) \cup (a_2, a_3)$ and < 0 on $(a_1, a_2) \cup (a_3, \infty)$. A similar statement holds for the three roots a'_1, a'_2, a'_3 of $D_2(w)$. Furthermore $a'_j = (\beta/\alpha)a_j$, $i \leq j \leq 3$, and $\alpha \leq a_3 \leq \beta \leq a'_3$, the equality signs holding if and only if $\alpha = \beta$.

In the formulas of Theorem 2.2, the constants $\sqrt{a_3 - 1}$, etc. are assumed positive and the functions $\sqrt{a_3 - z}$, $\sqrt{a_3 - w}$ are those analytic branches which are positive at the origin.

THEOREM 2.2. *Let*

$$(2.6) \quad \phi(z) = \frac{\sqrt{a_3 - z} + \sqrt{a_3 - 1}}{[\sqrt{a_3 - z} + \sqrt{a_3 - \alpha/\beta}] \cdot [\sqrt{a_3 - z} - \sqrt{a_3 - \alpha}]}, \quad |z| \leq 1,$$

$$(2.7) \quad \psi(w) = \frac{\sqrt{a'_3 - w} + \sqrt{a'_3 - 1}}{[\sqrt{a'_3 - w} + \sqrt{a'_3 - \beta/\alpha}] \cdot [\sqrt{a'_3 - w} - \sqrt{a'_3 - \beta}]}, \quad |w| \leq 1.$$

Then

$$(2.8) \quad P(z, w) = \frac{\beta - 1}{\beta} \frac{\phi(z)}{\phi(1)}, \quad |z| \leq 1,$$

$$(2.9) \quad P(0, w) = \frac{\alpha - 1}{\alpha} \frac{\psi(w)}{\psi(1)}, \quad |w| \leq 1.$$

For $\alpha = \beta$, (2.8), (2.9) become

$$(2.10) \quad P(z, 0) = P(0, z) = \frac{(\alpha - 1)^{3/2}}{\alpha(\alpha - z)^{1/2}}.$$

Formulas (2.8), (2.9) provide analytic continuations of $P(z, 0)$, $P(0, w)$ to the z - and w -planes minus the respective slits $[a_3, \infty]$, $[a'_3, \infty]$. If $\alpha < \beta$, then $P(z, 0)$ is analytic in the slit z -plane except for a simple pole at α , while $P(0, w)$ is analytic in the slit w -plane. If $\alpha = \beta$, then both $P(z, 0)$, $P(0, w)$ are analytic in the slit z - and w -planes.

$P(z, w)$ also has an analytic continuation beyond $|z|, |w| < 1$. It does not seem easy to obtain this fact directly from the formula for $P(z, w)$. The following is a simple proof. Let $P(z, 1) = \sum P_m z^m$, $P(1, w) = \sum Q_n w^n$, $|z|$ and $|w| \leq 1$, where $P_m = \sum_n p_{mn}$, $Q_n = \sum_m p_{mn}$. Letting $z = 1$, $w = 1$ respectively in (2.2) and using $P(1, 1) = \sum p_{mn} = 1$, we get

$$(2.11) \quad P(z, 1) = \frac{\alpha - 1}{\alpha - z}, \quad Q(1, w) = \frac{\beta - 1}{\beta - w}.$$

Hence

$$(2.12) \quad P_m = \frac{\alpha - 1}{\alpha^{m+1}}, \quad Q_n = \frac{\beta - 1}{\beta^{n+1}}$$

and

$$(2.13) \quad p_{mn} \leq \min [P_m, Q_n] \leq \sqrt{P_m Q_n} = \sqrt{\frac{\alpha-1}{\alpha}} \sqrt{\frac{\beta-1}{\beta}} \left(\frac{1}{\sqrt{\alpha}}\right)^m \left(\frac{1}{\sqrt{\beta}}\right)^n.$$

(2.13) shows that $P(z, w) = \sum p_{mn} z^m w^n$ converges absolutely for $|z| < \sqrt{\alpha}$, $|w| < \sqrt{\beta}$, which is the same as saying that it is analytic for these values of z and w .

We explain the technique used to derive Theorem 2.2 in some detail, as the encountered notions are employed throughout this paper. From (2.2) we conclude that $N(z, w) = 0$ on the portion $|z|, |w| \leq 1$ of the algebraic curve S with equation $Q(z, w) = 0$. We parametrize the algebraic curve, or equivalently, the Riemann surface S and reinterpret (2.2) in terms of the parametrization (Theorem 2.4). Solving $Q = 0$ for w , we get

$$(2.14) \quad w(z) = \frac{1 + \alpha + \beta \pm \sqrt{D_1(z)}}{2z^2}.$$

Thus S may be pictured as a two-sheeted Riemann surface over the z -plane with branch points at a_1, a_2, a_3, ∞ . S can be parametrized in the following manner. By the Schwartz–Christoffel transformation [6, Chap. 5] there exists a closed rectangle R_0 with vertices $0, \tilde{\omega}, \tilde{\omega} + \omega, \omega$, where $\tilde{\omega}, \omega/i > 0$, and a one-to-one mapping $t(z)$ from H , the closed upper half of the z -plane, to R_0 such that: $t(z)$ is continuous on H , $t(z)$ is analytic on $\text{Int } H = \text{interior of } H$, $t(z)$ maps a_1, a_2, a_3, ∞ to the four vertices of R_0 in the order listed. Let $z = z(t)$ be the inverse mapping. By repeated use of the reflection principle, $z(t)$ can be extended as an even elliptic function over the entire t -plane with periods $2\tilde{\omega}, 2\omega$.

For $z \in H$, let $\sqrt{D_1(z)}$ be that square root which is continuous on H and $\sqrt{D_1(0)} = \alpha$. In (2.14) choose the plus sign and let $w(t) = w(z(t))$, $t \in \text{Int } R_0$. $w(t)$ is analytic on $\text{Int } R_0$ and can also be extended to the entire t -plane as an elliptic function with periods $2\tilde{\omega}, 2\omega$. As t varies over the t -plane, $\phi(t) = (z(t), w(t))$ varies over all of S . It is readily checked that $\phi(t_1) = \phi(t_2)$ if and only if $t_2 = t_1 + 2m\tilde{\omega} + 2n\omega$, m and n arbitrary integers. Thus S is parametrized by the pair of elliptic functions $z = z(t)$, $w = w(t)$.

We shall need to know the values of $z(t)$, $w(t)$ at certain points of the fundamental rectangle $R = \{t: -\tilde{\omega} < \text{Re } t \leq \tilde{\omega}, -\text{Im } \omega < \text{Im } t \leq \text{Im } \omega\}$. For any two complex numbers a, b we use the symbols $[a, b]$, (a, b) , $[a, b)$, $(a, b]$ to denote respectively the closed, open, and half-open line segments directed from a to b , $[a, b)$ containing a and $(a, b]$ containing b . From the mapping properties of $z(t)$, it readily follows that there exist unique points m and p, q lying respectively on $(0, \omega)$, $(\tilde{\omega}, \tilde{\omega} + \omega]$ such that $z(m) = 0$, $z(p) = 1$, $z(q) = \alpha$. Furthermore p precedes q on $(\tilde{\omega}, \tilde{\omega} + \omega]$, which we denote by $p < q$. The derivation of the following theorem follows from (2.2) and standard theorems on elliptic functions [6, Chap. 8].

THEOREM 2.3. (i) *The zeros and poles of $z(t)$ in R are respectively $\{-m, m\}$, $\{\omega, \omega\}$ (i.e. ω is a double pole of $z(t)$).*

(ii) *The zeros and poles of $w(t)$ in R are respectively $\{m, \omega\}$, $\{-m, -m\}$.*

(iii) *$m = \omega/3$.*

(iv) *$w(t) = (\beta/\alpha)z(t-2m)$. As $z(-t) = z(t)$, we obtain $w(4m-t) = w(t)$.*

(v) *The zeros and poles of $z'(t)$ in R are $\{0, \tilde{\omega}, \tilde{\omega} + \omega\}$, $\{\omega, \omega, \omega\}$. Those of $w'(t)$ are $\{2m, \tilde{\omega} + 2m, \tilde{\omega} - m\}$, $\{-m, -m, -m\}$.*

(vi) *$(p+q)/2 = \tilde{\omega} + 2m$ and $\tilde{\omega} + m < p < \tilde{\omega} + 2m < \tilde{\omega} + \omega$, with equality holding if and only if $\alpha = \beta$.*

(vii) At the points $\{p-4m, q-4m, p, q\}$, $z(t)$ and $w(t)$ assume the following values.

t	$p-4m$	$q-4m$	p	q
$z(t)$	α	1	1	α
$w(t)$	$\frac{\beta}{\alpha}$	β	1	1

The positions of the points p, q, m in R are indicated in Fig. 1. From (iv) we obtain the mapping properties of $w(t)$ which are similar to those of $z(t)$. Let R'_0 be the rectangle with vertices $2m, \tilde{\omega}+2m, \tilde{\omega}+5m, 5m$. R'_0 is R_0 translated by $2m$. $w(t)$ maps R'_0 onto the upper half of the w -plane, taking the four vertices in the order listed to a'_1, a'_2, a'_3, ω .

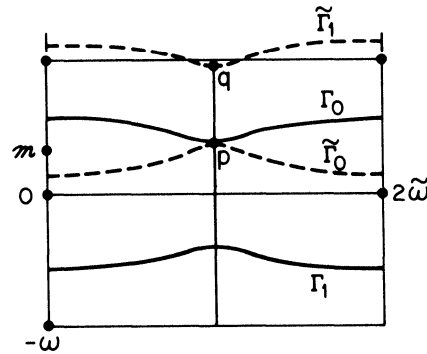


FIG. 1

Let $D_z = \{t: |z(t)| < 1\}$, $D_w = \{t: |w(t)| < 1\}$. The closures of these sets are $\bar{D}_z = \{t: |z(t)| \leq 1\}$, $\bar{D}_w = \{t: |w(t)| \leq 1\}$. From the mapping properties of $z(t)$ it follows that D_z consists of an infinite number of simply connected components which are translates by integral multiples of 2ω of the component D_z containing 0. A similar statement applies to D_w , D_w being the component containing $2m$. Let Γ_1, Γ_0 be the lower and upper boundaries of D_z and $\tilde{\Gamma}_0, \tilde{\Gamma}_1$ the lower and upper boundaries of D_w . The mapping properties of $z(t)$, $w(t)$ imply that: D_z, D_w are invariant under $t \rightarrow t + 2\tilde{\omega}$, D_z is symmetric with respect to the real and imaginary axes, D_w is symmetric with respect to the horizontal line through $2m$ and the imaginary axis. Thus the curves $\Gamma_j, \tilde{\Gamma}_j$ are completely described by their portions in the strip $\{t: 0 \leq \operatorname{Re} t \leq \tilde{\omega}\}$. In Fig. 1, the curves are sketched for $0 \leq \operatorname{Re} t \leq 2\tilde{\omega}$. (A precise description of D_z, D_w is given in [2, Thm. 4.1].) Observe that $\Gamma_0, \tilde{\Gamma}_0$ intersect only at the points $p + 2m\tilde{\omega}, m$ an arbitrary integer.

Let $A(t) = P(z(t), 0)$, $B(t) = P(0, w(t))$. $A(t), B(t)$ are respectively analytic in D_z, D_w and continuous in $\bar{D}_z, \bar{D}_w$. The functional equation (2.2) for $P(z, w)$ converts into the following conditions for $A(t), B(t)$.

THEOREM 2.4.

$$(2.15) \quad \begin{cases} A(-t) = A(t), & t \in \bar{D}_z, \\ B(4m - t) = B(t), & t \in \bar{D}_w, \\ B(t) = R(t)A(t), & t \in \bar{D}_z \cap \bar{D}_w \end{cases}$$

where

$$(2.16) \quad R(t) = -\frac{\beta}{\alpha} \frac{z(t)[w(t)-1]}{w(t)[z(t)-1]}.$$

As shown in [2], one can use (2.15) to prove that $A(t)$, $B(t)$ can be extended to the entire t -plane as elliptic functions with periods $2\tilde{\omega}$, 4ω . Equations (2.15) then hold for all t by analytic continuation. Knowledge of the zeros and poles of $A(t)$, $B(t)$ leads to their determination and to the formulas for $P(z, 0)$, $P(0, w)$ given in Theorem 2.2.

3. Integral formulas. We need the following facts concerning the zeros of $Q(z, w)$.

LEMMA. *Let $|z| = 1 + \varepsilon$. For $\varepsilon > 0$ sufficiently small, $Q(z, w) = 0$ has two roots $w_1(z)$, $w_2(z)$ satisfying $|w_1(z)| < 1$, $|w_2(z)| > 1 + \varepsilon$. A similar statement holds upon reversing z and w .*

Proof. Let $|z| = 1 + \varepsilon$, $|w| = 1$. Then

$$(3.1) \quad |(1 + \alpha + \beta)zw| = f(\varepsilon) = (1 + \alpha + \beta)(1 + \varepsilon),$$

$$(3.2) \quad |\alpha w + \beta z + z^2 w^2| \leq g(\varepsilon) = \alpha + \beta(1 + \varepsilon) + (1 + \varepsilon)^2.$$

$f(0) = g(0)$ and $f'(0) > g'(0)$, so that $f(\varepsilon) > g(\varepsilon)$ for $\varepsilon > 0$ sufficiently small. We conclude from Rouché's theorem that, for fixed z , $Q(z, w) = 0$ has two roots $w_1(z)$, $w_2(z)$ with $|w_1(z)| < 1 < |w_2(z)|$. Since $w_2(z) = \beta / zw_1(z)$, we have $|w_2(z)| > \beta / (1 + \varepsilon) > 1 + \varepsilon$ for $\varepsilon > 0$ sufficiently small. The same proof works upon reversing z and w .

From now on Γ_1 , $\tilde{\Gamma}_1$ denote those portions of the curves contained in the strip $0 \leq \operatorname{Re} t \leq 2\tilde{\omega}$. We orient Γ_1 , $\tilde{\Gamma}_1$ from left to right, i.e. with initial point on $\operatorname{Re} t = 0$ and end point on $\operatorname{Re} t = 2\tilde{\omega}$.

THEOREM 3.1. *Let $z = z(t)$, $z' = dz/dt$, etc. Then*

$$(3.3) \quad p_{mn} = \frac{1}{2\pi i} \left[\int_{\tilde{\Gamma}_1} - \int_{\Gamma_1} \right] \frac{\beta(w-1)Az'}{z^m w^{n+1} \partial Q / \partial w} dt = \frac{1}{2\pi i} \left[\int_{\tilde{\Gamma}_1} - \int_{\Gamma_1} \right] \frac{\alpha(z-1)Bw'}{z^{m+1} w^n \partial Q / \partial z} dt.$$

We remark that (3.3) can be simplified slightly. We have $(\partial Q / \partial w)^2 = D_1(z)$ when $z = z(t)$, $w = w(t)$. From this we readily conclude that z' , $\partial Q / \partial w$ have the same zeros and poles. Hence $z' / (\partial Q / \partial w) = c$, where c is a nonzero constant.

Proof. Choose ε_0 so small that $P(z, w)$ is analytic for $|z|$, $|w| < 1 + \varepsilon_0$, and the lemma holds for $0 < \varepsilon < \varepsilon_0$. From the formula for $P(z, w)$ and Cauchy's formula for p_{mn} , we have, for $1 < r < 1 + \varepsilon_0$,

$$(3.4) \quad p_{mn} = \frac{1}{2\pi i} \int_{|z|=r} \frac{\beta P(z, 0)}{z^m} \left[\frac{1}{2\pi i} \int_{|w|=r} \frac{w-1}{Qw^{n+1}} dw \right] dz \\ + \frac{1}{2\pi i} \int_{|w|=r} \frac{\alpha P(0, w)}{w^n} \left[\frac{1}{2\pi i} \int_{|z|=r} \frac{z-1}{Qz^{n+1}} dz \right] dw = I_1 + I_2,$$

the circles $|z| = r$, $|w| = r$ being oriented positively.

Applying the residue theorem, we get, for $R > r$,

$$(3.5) \quad \frac{1}{2\pi i} \int_{|w|=r} \frac{w-1}{Qw^{n+1}} dw = -\frac{w_2(z)-1}{Q_w(z, w_2(z))[w_2(z)]^{n+1}} + \frac{1}{2\pi i} \int_{|w|=R} \frac{w-1}{Qw^{n+1}} dw.$$

We have

$$(3.6) \quad \int_{|w|=R} \frac{w-1}{Qw^{n+1}} dw = O\left(\frac{1}{R^{n+1}}\right) \quad \text{as } R \rightarrow \infty.$$

Hence, letting first $R \rightarrow \infty$ and then $r \rightarrow 1$, we get

$$(3.7) \quad I_1 = \frac{-1}{2\pi i} \int_{|z|=1} \frac{\beta P(z, 0)(w_2(z)-1)}{z^m (w_2(z))^{n+1} (\partial Q / \partial w)(z, w_2(z))} dz.$$

$z = z(t)$ maps Γ_1 onto $|z| = 1$ with positive orientation, and $w(t) = w_2(z(t))$. Hence

$$(3.8) \quad I_1 = -\frac{1}{2\pi i} \int_{\Gamma_1} \frac{\beta(w-1)Az'}{z^m w^{n+1} \partial Q / \partial w} dt.$$

Similar reasoning applies to I_2 , except that $w = w(t)$ maps $\tilde{\Gamma}_1$ onto $|w| = 1$ oriented negatively. Hence

$$(3.9) \quad I_2 = \frac{1}{2\pi i} \int_{\tilde{\Gamma}_1} \frac{\alpha(z-1)Bw'}{z^{m+1} w^n \partial Q / \partial z} dt.$$

Differentiating $Q(z, w) = 0$ with respect to t , we get

$$\frac{w'}{\partial Q / \partial z} = -\frac{z'}{\partial Q / \partial w}.$$

By Theorem 2.4 $\beta z(w-1)A + \alpha w(z-1)B = 0$ for all t . (3.3) follows from (3.4), (3.8), (3.9).

4. Saddle points. We shall obtain the asymptotics of p_{mn} along the lines $n/m = \theta$, $0 \leq \theta < \infty$. Thus the term $(z^m w^n)^{-1}$ in formula (3.3) is to be replaced by u^m , where $u(t, \theta) = [z(t)w^\theta(t)]^{-1}$.

The function $w^\theta(t)$ is multiple valued and we must specify the chosen branch. Let $S = \{t: 0 < \operatorname{Re} t < 2\tilde{\omega}\}$. $w(t) > 0$ for $\operatorname{Re} t = \omega$ and $w(t) \neq 0, \infty$ on S . For $0 \leq \theta < \infty$, let $w^\theta(t)$ be the branch which is positive on $\operatorname{Re} t = \omega$ and analytic in S . Extend $w^\theta(t)$ to the closure $\bar{S} = \{t: 0 \leq \operatorname{Re} t \leq 2\tilde{\omega}\}$ by taking limits. Thus $u(t, \theta)$ is defined for $0 \leq \theta < \infty$, $t \in \bar{S}$. $u(t, \theta)$ is analytic in t except at $(2j+1)m + k\tilde{\omega}$, j any integer and $k = 0, 1$, and $u(t+2\omega) = u(t)$.

For given θ , the saddle points of u are defined as the zeros of $u'(t, \theta)$ in S , u' denoting the derivative of u with respect to t . We have

$$(4.1) \quad u' = -u \frac{w'}{w} (h + \theta), \quad h = \frac{wz'}{zw'} = \frac{-w \partial Q / \partial w}{z \partial Q / \partial z}.$$

In S u has no zeros and the zeros of w'/w are poles of $h + \theta$. It follows from (4.1) that the saddle points of u are identical with the zeros of $h + \theta$ in S . We show how the zeros of $h + \theta$ vary with θ , $0 \leq \theta \leq \infty$ (for $\theta = \infty$, a zero of $h + \theta$ is interpreted as a pole of h).

We use the following terminology. Let $f(t)$ be an elliptic function with periods $2\tilde{\omega}, 2\omega$. Let $a_1, \dots, a_k$ be the zeros of $f(t)$ in R and $\alpha_1, \dots, \alpha_k$ any set of numbers satisfying $\alpha_i - a_i = 2m_i\tilde{\omega} + 2n_i\omega$, m_i and n_i integers, $1 \leq i \leq k$. $\alpha_1, \dots, \alpha_k$ are then said to form a complete set of zeros of $f(t)$. Similar terminology applies to the poles of $f(t)$.

THEOREM 4.1. i) Let $I_1 = [\tilde{\omega} - \omega, \tilde{\omega} - m]$, $I_2 = [\tilde{\omega}, \tilde{\omega} + 2m]$, $I_3 = [0, -4m]$. For $0 \leq \theta \leq \infty$ and $1 \leq j \leq 3$, there exists a unique $t_j(\theta) \in I_j$ such that the points $t_j(\theta)$ form a complete set of zeros of $h(t) + \theta$. As θ increases from 0 to ∞ , $t_j(\theta)$ moves in one direction from initial to end point of I_j .

ii) $(t_j(\theta) + t_j(1/\theta))/2 = p_j$, where p_j is the mid-point of I_j . Hence $t_j(\theta)$ lies in the first half of I_j for $0 \leq \theta \leq 1$, and in the second half for $1 \leq \theta \leq \infty$.

The contents of Theorem 4.1 are illustrated in Fig. 2, the arrow on I_j indicating the direction in which $t_j(\theta)$ moves as θ increases.

Proof. i) Parametrize I_1 by $t = \tilde{\omega} - \omega + 2m\tau$, $0 \leq \tau \leq 1$. From the mapping properties of z, w we get $z, w, (dw/d\tau), -dz/d\tau > 0$, $0 < \tau < 1$. Hence

$$h = \frac{w dz/d\tau}{z dw/d\tau} < 0, \quad 0 < \tau < 1.$$

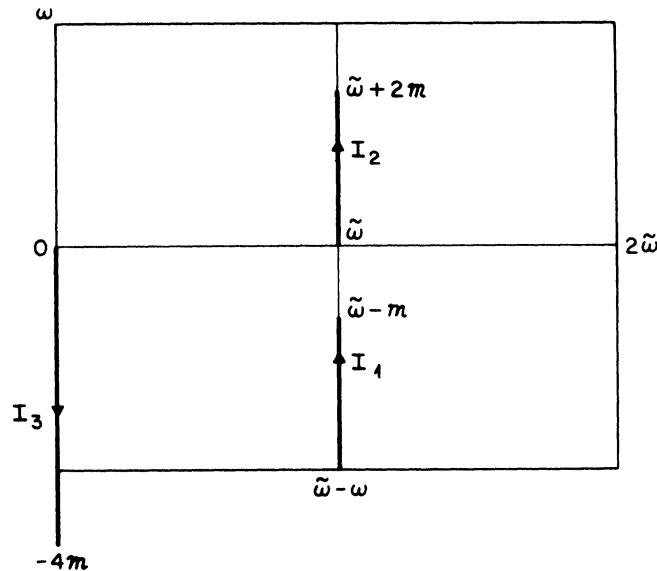


FIG. 2

Similar considerations show that $h < 0$ on the interiors of I_2, I_3 . From Theorem 2.3 we find that the initial points of I_j , $1 \leq j \leq 3$, give a complete set of zeros of h , and the end points a complete set of poles of h . Hence h assumes every value three times in any given fundamental rectangle. It follows that $h(t)$ decreases from 0 to $-\infty$ as t moves from initial to end point of I_j , so that $h(t) + \theta = 0$ has one solution in each I_j .

ii) Let $(t + \tilde{t}_j)/2 = p_j$. $w(t) = (\beta/\alpha)z(t - 2m)$ and $2p_j = 2m + 2m_j\tilde{\omega} + 2n_j\omega$, with m_j and n_j integers. Hence $w(t) = (\beta/\alpha)z(\tilde{t}_j)$, from which

$$(4.2) \quad h(t) = -\frac{z'}{z}(t) \bigg/ \frac{z'}{z}(\tilde{t}_j).$$

(4.2) implies $h(\tilde{t}_j) = 1/h(t)$, which is Theorem 4.1ii).

In the sequel, the only saddle point of interest to us is $t_\theta = t_1(\theta)$, as the steepest descent curve defined in § 5 passes through t_θ .

THEOREM 4.2. *Let $0 \leq \theta < \infty$, $t \in I_1$. For fixed θ , $u(t, \theta)$ attains an absolute minimum at t_θ and $u''(t_\theta, \theta) < 0$.*

Proof. Parametrize I_1 as above and let $t_\theta = \tilde{\omega} - \omega + 2m\tau_\theta$. We have

$$(4.3) \quad \frac{du}{d\tau} = \frac{-u(dw/d\tau)(h + \theta)}{w}, \quad 0 \leq \tau \leq 1,$$

$$(4.4) \quad \frac{d^2u}{d\tau^2} = \frac{-u}{w} \frac{dw}{d\tau} \frac{dh}{d\tau}, \quad \tau = \tau_\theta.$$

Use the mapping properties of z, w and Theorem 4.1 to conclude from (4.3) that $du/d\tau < 0$ for $\tau < \tau_\theta$ and > 0 for $\tau > \tau_\theta$. Hence u has an absolute minimum at τ_θ . Similarly, we conclude from (4.3) that $d^2u/d\tau^2 > 0$ at τ_θ , which is the same as $u''(t_\theta, \theta) < 0$.

5. Steepest descent curve. The proof of Theorem 5.1 depends on the mapping properties of $u = u(t, \theta)$ and is deferred to the appendix. We use it here to establish the existence of a steepest descent curve Γ_θ^* for $u(t, \theta)$, $0 \leq \theta < \infty$.

THEOREM 5.1. Let R_1 be the closed rectangle with vertices $-\omega, \tilde{\omega} - \omega, \tilde{\omega} - 2m, -2m$. (R_1 is the shaded rectangle in Fig. 3.) Let $0 < \theta < 1$ and $u_\theta = u(t_\theta, \theta)$. For $0 < u < u_\theta$, the equation $u(t, \theta) = u$ has precisely one solution in the interior of R_1 which we denote by $t = t(u, \theta)$. $t(u, \theta)$ is analytic on $(0, u_\theta)$ and has a continuous extension to $[0, u_\theta]$ by letting $t(0, \theta) = -\omega$, $t(u_\theta, \theta) = t_\theta$.

For $0 < \theta < 1$, let $\Gamma_{\theta, \ell}^*$ be the curve with equation $t = t(u, \theta)$, $0 \leq u \leq u_\theta$, and let $\Gamma_{\theta, r}^*$ be the reflection of $\Gamma_{\theta, \ell}^*$ in γ_2 , the right side of R_1 . Let $\Gamma_\theta^* = \Gamma_{\theta, \ell}^* \cup \Gamma_{\theta, r}^*$. For $1 < \theta < \infty$, define Γ_θ^* to be the reflection of $\Gamma_{1/\theta}^*$ in γ_3 , the upper side of R_1 . For $\theta = 0$, let $\Gamma_0^* = [-\omega, 2\tilde{\omega} - \omega]$, $\Gamma_1^* = [-2m, 2\tilde{\omega} - 2m]$.

By Theorems 4.1, 5.1, Γ_θ^* passes through t_θ for $0 \leq \theta < \infty$. Γ_θ^* is oriented from left to right and is sketched in Fig. 3 for various θ .

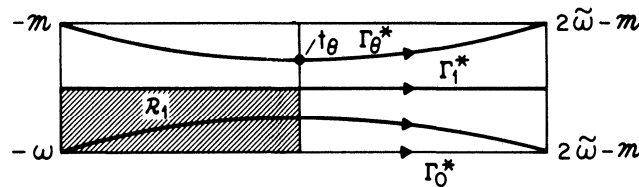


FIG. 3

Let $0 \leq \theta < 1$. As t traverses $\Gamma_{\theta, \ell}^*$ from t_θ to $-\omega$, $u(t, \theta)$ decreases from u_θ to 0. Since $u(t, \theta)$ is real on γ_2 , the reflection principle gives $u(t', \theta) = u(t, \theta)$, where t' is the reflection of t in γ_2 . Hence $u(t, \theta)$ also decreases from u_θ to 0 as t traverses $\Gamma_{\theta, r}^*$ from t_θ to $2\tilde{\omega} - \omega$. This is the reason for naming Γ_θ^* a steepest descent curve for $u(t, \theta)$.

The steepest descent property of Γ_θ^* persists for $1 \leq \theta < \infty$. $t'' = \bar{t} - 4m$ is the reflection of t in γ_3 . Since $z(t)$ is real t , we have $z(\bar{t}) = \overline{z(t)}$. This implies $w(t) = (\beta/\alpha)z(t'')$, $z(t) = (\alpha/\beta)\overline{w(t'')}$, so that

$$(5.1) \quad u(t, \theta) = \left(\frac{\beta}{\alpha}\right)^{1-\theta} u^\theta\left(t'', \frac{1}{\theta}\right), \quad \theta > 0,$$

where $(\beta/\alpha)^{1-\theta} > 0$, $u^\theta(t, 1/\theta) > 0$ for $\text{Re } t = \tilde{\omega}$, and u^θ is extended to S by analytic continuation. The desired property of Γ_θ^* , $1 < \theta < \infty$, follows from (5.1). In Lemma 1 of the appendix we show that $u(t, 1) = z(t + 2m)/\alpha$, which implies that $u(t, 1)$ decreases from a_2/α to a_1/α as t traverses either half of Γ_1^* away from t_1 . Thus Γ_1^* also has the desired property.

THEOREM 4.2. Γ_θ^* is rectifiable for $0 \leq \theta < \infty$.

Proof. It suffices to consider $0 < \theta < 1$. The length $\ell(\Gamma_\theta^*)$ of Γ_θ^* is given by

$$(5.2) \quad \ell(\Gamma_\theta^*) = \int_0^{u_\theta} \left| \frac{dt}{du} \right| du = \int_0^{u_\theta} \frac{du}{|u'|}.$$

We prove that the integral is finite. In the argument which follows, t is always in R_1 . For fixed θ , $u(t, \theta)$ is analytic in R_1 except at $-\omega$, where $z(t)$, $w(t)$ have respectively a double pole and a simple zero. Thus for t sufficiently near $-\omega$

$$(5.3) \quad u(t, \theta) = (t + \omega)^{2-\theta} g(t)$$

where $g(t)$ is analytic with $g(-\omega) \neq 0$, $(t + \omega)^{2-\theta} > 0$ for $t + \omega > 0$, and $(t + \omega)^{2-\theta}$ is determined elsewhere by analytic continuation. The zeros of $u'(t, \theta)$ are at $-\omega$ and t_θ . From the expansions for u near these points, we readily conclude that there exists neighborhoods N_1 , N_2 about $-\omega$, t_θ and a constant $C > 0$ such that

$$(5.4) \quad |u'| \geq C|u|^{1/2} (t \in N_1), \quad |u'| \geq C|u - u_\theta|^{1/2} (t \in N_2), \quad |u'| \geq C (t \in N \cup N_2).$$

Hence

$$(5.5) \quad \ell(\Gamma_\theta^*) \leq \frac{1}{C} \int_0^{u_\theta} \max \left[1, \frac{1}{\sqrt{u}}, \frac{1}{\sqrt{u_\theta - u}} \right] du < \infty.$$

We remark that a slight refinement of the proof gives a uniform bound for $\ell(\Gamma_\theta^*)$, $0 \leq \theta < \infty$.

6. Asymptotic formulas.

THEOREM 6.1. Let $m > 0$, $n/m = \theta$, and

$$(6.1) \quad \Phi(t) = \frac{B(w-1)z'}{w \partial Q / \partial w} \cdot [A(t+2\omega) - A(t)] = \frac{\alpha(z-1)z'}{z \partial Q / \partial z} [B(t+2\omega) - B(t)].$$

If $\beta > \alpha$, then

$$(6.2) \quad p_{mn} = \begin{cases} \frac{1}{2\pi i} \int_{\Gamma_\theta^*} \Phi u^m dt + \frac{(\alpha-1)(\beta-\alpha)}{\alpha^{m-n+1}\beta^{n+1}}, & 0 \leq \theta < \frac{\beta-\alpha}{\beta-1}, \\ \frac{1}{2\pi i} \int_{\Gamma_\theta^*} \Phi u^m dt, & \frac{\beta-\alpha}{\beta-1} < \theta < \infty. \end{cases}$$

If $\beta = \alpha$, then the second formula of (6.2) holds for $0 \leq \theta < \infty$. Observe that we have omitted the case $\theta = (\beta - \alpha) / (\beta - 1) > 0$. We show below that in this case $\Phi(t)$ has a pole at t_θ , so that (6.2) is no longer valid. The second formula of (6.2) can be shown to hold if we indent Γ_θ^* near t_θ , so that the indented curve lies above t_θ .

Proof. (6.3) is derived from (3.3). Let

$$F = \frac{\beta(w-1)Az'}{z^m w^{n+1} \partial Q / \partial w}.$$

Let Γ be the positively oriented curve consisting of Γ_θ^* , $-\Gamma_1$, and the two vertical segments joining respectively the initial and end points of Γ_θ^* , Γ_1 . $F(t)$ is analytic on Γ . Since $F(t+2\omega) = F(t)$, the contribution of the vertical segments to $\int_\Gamma F(t) dt$ cancel out, so that

$$(6.3) \quad \int_\Gamma F(t) dt = \int_{\Gamma_\theta^*} F(t) dt - \int_{\Gamma_1} F(t) dt.$$

The only pole of $F(t)$ surrounded by Γ is the simple pole $p-4m$, and this only in case $\beta > \alpha$ and Γ_θ^* lies below $p-4m$ (when $\alpha = \beta$, $F(t)$ is analytic at $p-4m$). Since $z(p-4m) = \alpha$, $w(p-4m) = \beta/\alpha$, we conclude from (4.1) that $h(p-4m) = (\alpha - \beta) / (\beta - 1)$, i.e. $p-4m = t_{((\beta-\alpha)/(\beta-1))}$. Hence Γ_θ^* lies below $p-4m$ if and only if $0 \leq \theta < (\beta - \alpha) / (\beta - 1)$. Γ is depicted in Fig. 4 for $0 \leq \theta < (\beta - \alpha) / (\beta - 1)$.

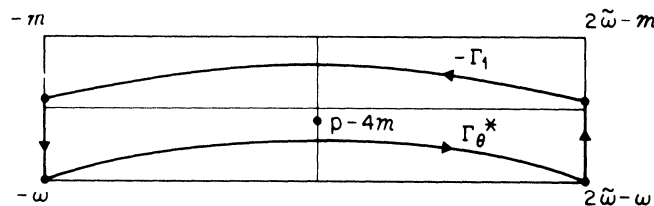


FIG. 4

In (7.9) of [2] we show that, for $\alpha < \beta$, $P(z, 0)$ has a simple pole at α with residue $(\alpha - 1)(\alpha - \beta)/\beta$. Since $A(t) = P(z(t), 0)$, we can compute the residue of $F(t)$ at $p - 4m$. The residue theorem gives, for $\alpha < \beta$,

$$(6.4) \quad \int_{\Gamma_1} F(t) dt = \begin{cases} \int_{\Gamma_\theta^*} F(t) dt - 2\pi i \frac{(\alpha - 1)(\beta - \alpha)}{\alpha^{m-n+1}\beta^{n+1}}, & 0 \leq \theta < \frac{\beta - \alpha}{\beta - 1}, \\ \int_{\Gamma_\theta^*} F(t) dt, & \frac{\beta - \alpha}{\beta - 1} < \theta < \infty. \end{cases}$$

Similarly, for $\alpha = \beta$, the second formula of (6.4) holds.

$\int_{\tilde{\Gamma}_1} F(t) dt$ is treated in analogous manner. In deforming $\tilde{\Gamma}_1$ to $\Gamma_\theta^* + 2\omega$ no poles of $F(t)$ are encountered. Hence

$$(6.5) \quad \int_{\tilde{\Gamma}_1} F(t) dt = \int_{\Gamma_\theta^* + 2\omega} F(t) dt = \int_{\Gamma_\theta^*} F(t + 2\omega) dt.$$

(6.2) follows from (3.3), (6.4), (6.5).

The asymptotics of p_{mn} are derived from (6.2). We introduce some notation. Let $z_\theta = z(t_\theta)$, $w = w(t_\theta)$, $0 \leq \theta < \infty$. (z_θ, w_θ) traces out the curve ρ with equation $Q(z, w) = 0$, $\sqrt{\alpha/a'_3} \leq z \leq a_3$ and $\sqrt{\beta/a'_3} \leq w \leq a'_3$. ρ is sketched in Fig. 5, the arrow indicating the direction in which (z_θ, w_θ) moves as θ increases. For given θ and along ρ , we may think of u as either a function of z or w . u is then analytic in z except at a_3 and in w except at a'_3 .

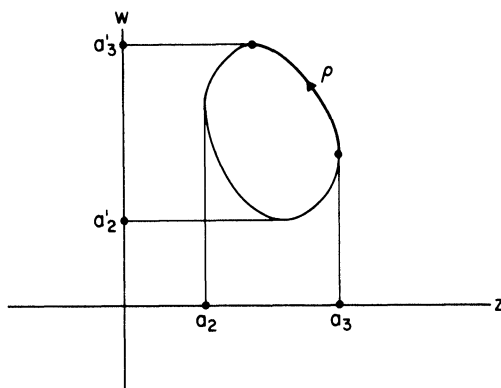


FIG. 5

Let $\phi_-(z)$ the function obtained from $\phi(z)$, defined in (2.6), upon replacing $\sqrt{a_3 - z}$ by $-\sqrt{a_3 - z}$. All square roots appearing in the statement and proof of Theorem 6.2 are assumed positive.

THEOREM 6.2. *Let $m > 0$, $n/m = \theta$. If $\alpha < \beta$, then*

i)

$$(6.6) \quad p_{mn} \sim \frac{(\alpha - 1)(\beta - \alpha)}{\alpha^{m-n+1}\beta^{n+1}} \quad \text{as } m \rightarrow \infty \text{ uniformly for } 0 \leq \theta \leq a, \\ \text{whenever } 0 \leq a < \frac{\beta - \alpha}{\beta - 1}.$$

ii)

$$(6.7) \quad p_{mn} \sim \frac{(\alpha - 1)(\beta - \alpha)}{2\alpha^{m-n+1}\beta^{n+1}} \quad \text{as } m \rightarrow \infty \text{ for } \theta = \frac{\beta - \alpha}{\beta - 1} = \text{rational number}.$$

iii)

$$(6.8) \quad p_{mn} \sim \frac{C_\theta}{m^{1/2} z_\theta^m w_\theta^n} \quad \text{as } m \rightarrow \infty \text{ uniformly for } a \leq \theta \leq b,$$

$$\text{whenever } \frac{\beta - \alpha}{\beta - 1} < a < b < \infty,$$

where

$$(6.9) \quad C_\theta = \frac{(\beta - 1)(w_\theta - 1)}{w_\theta(\partial Q / \partial w)(z_\theta, w_\theta)} \cdot \frac{\phi_-(z_\theta) - \phi(z_\theta)}{\phi(1)} \cdot \left(\frac{u_\theta}{2\pi(\partial^2 u / \partial z^2)(z_\theta, \theta)} \right)^{1/2}.$$

If $\alpha = \beta$, then

iv)

$$(6.10) \quad p_{mn} \sim \frac{D_\theta}{m^{1/2} z_\theta^m w_\theta^n} \quad \text{as } m \rightarrow \infty \text{ uniformly for } 0 \leq \theta \leq a, \text{ whenever } 0 \leq a < \infty,$$

where

$$(6.11) \quad D_\theta = \frac{-2(\alpha - 1)^{3/2}(z_\theta - 1)}{z_\theta(\partial Q / \partial z)(z_\theta, w_\theta)(\alpha - w_\theta)^{1/2}} \cdot \left(\frac{u_\theta}{2\pi(\partial^2 u / \partial w^2)(w_\theta, \theta)} \right)^{1/2}.$$

Remark. Theorem 6.2 states that two asymptotic formulas prevail for p_{mn} , depending on whether $n/m < (\beta - \alpha)/(\beta - 1)$ or $n/m > (\beta - \alpha)/(\beta - 1)$. Analytically, this stems from the fact that (6.2) contains two formulas for p_{mn} . We have no heuristic explanation of this phenomenon, and no physical interpretation of the constant $(\beta - \alpha)/(\beta - 1)$.

Proof. Γ_θ^* is a steepest descent curve for $u(t, \theta)$, and the asymptotic behavior of p_{mn} is obtained from (6.2) by Laplace's method [1, Chap. 4]. To illustrate, we derive iii). The derivations of the remaining parts of the theorem are analogous and we omit them. Let u serve as parameter on Γ_θ^* . By Theorems 4.1, 4.2, for t near t_θ ,

$$(6.12) \quad u = u_\theta + \frac{u''(t_\theta, \theta)}{2} (t - t_\theta)^2 + \cdots, \quad u''(t_\theta, \theta) < 0.$$

Inverting the series and differentiating, we obtain for $u < u_\theta$ and u near u_θ ,

$$(6.13) \quad \Phi \frac{dt}{du} = \pm \frac{\Phi(t_\theta)}{\sqrt{-2u''(t_\theta, \theta)}} \cdot \frac{1}{\sqrt{u_\theta - u}} + \sum_{n=0}^{\infty} a_n (\pm \sqrt{u_\theta - u})^n$$

where the a_n 's depend on θ , the $+$ sign is chosen along $\Gamma_{\theta, \ell}^*$, and the $-$ sign along $\Gamma_{\theta, r}^*$. Laplace's method for obtaining the asymptotics of p_{mn} consists of replacing $\Phi(dt/du)$ in (6.2) by the first term of expansion (6.13), i.e.

$$(6.14) \quad p_{mn} \sim \frac{\Phi(t_\theta)}{\pi i \sqrt{-2u''(t_\theta, \theta)}} \int_0^{u_\theta} \frac{u^m du}{\sqrt{u_\theta - u}} = \frac{\Phi(t_\theta) u_\theta^{m+1/2}}{\pi i \sqrt{-2u''(t_\theta, \theta)}} \int_0^1 \frac{v^m}{\sqrt{1-v}} dv.$$

The last integral in (6.14) equals the beta function $B(m+1, \frac{1}{2}) \sim \sqrt{\pi/m}$ [3, pp. 257-258]. Thus

$$(6.15) \quad p_{mn} \sim -i \Phi(t_\theta) u_\theta^m \sqrt{\frac{-u_\theta}{2\pi u''(t_\theta, \theta) m}}.$$

Now $u''(t_\theta, \theta) = (\partial^2 u / \partial z^2)(z_\theta, \theta)(z'(t_\theta))^2$, and the mapping properties of $z(t)$ imply that $\iota z'(t_\theta) < 0$. Hence

$$(6.16) \quad \sqrt{-2u''(t_\theta, \theta)} = -\iota z'(t_\theta) \sqrt{2 \frac{\partial^2 u}{\partial z^2}(z_\theta, \theta)}.$$

The argument of [2, § 6] used to prove Theorem 2.2 shows that

$$A(t) = \begin{cases} \frac{\beta-1}{\beta} \frac{\phi(z(t))}{\phi(1)} & \text{if } -\text{Im } \omega < \text{Im } t < \text{Im } \omega, \\ \frac{\beta-1}{\beta} \frac{\phi_-(z(t))}{\phi(1)} & \text{if } \text{Im } \omega < \text{Im } t < 3 \text{Im } \omega. \end{cases}$$

Hence

$$A(t_\theta) = \frac{\beta-1}{\beta} \frac{\phi(z_\theta)}{\phi(1)}, \quad A(t_\theta + 2\omega) = \frac{\beta-1}{\beta} \frac{\phi_-(z_\theta)}{\phi(1)}.$$

(6.1) then gives

$$(6.17) \quad \Phi(t_\theta) = \frac{(\beta-1)(w_\theta-1)z'(t_\theta)}{w_\theta(\partial Q / \partial w)(z_\theta, w_\theta)} \cdot \frac{\phi_-(z_\theta) - \phi(z_\theta)}{\phi(1)}.$$

iii) follows from (6.15)–(6.17). We have derived it for fixed θ , and a check of all involved estimates reveals that the asymptotics are uniform on any closed subinterval of $((\beta-\alpha)/(\beta-1), \infty)$.

7. Proof of Theorem 1. We first prove

LEMMA. *Let $\alpha < \beta$. Then*

$$(7.1) \quad P(z, 0) = \frac{r_{-1}}{z-\alpha} + r_0 + \cdots, \quad z \text{ near } \alpha$$

with

$$(7.2) \quad r_{-1} = \frac{(\alpha-1)(\alpha-\beta)}{\beta}, \quad \frac{\beta r_0}{\alpha-1} + \frac{\partial P}{\partial w}(0, 1) = \frac{\beta-1}{\beta-\alpha}.$$

Proof. By Theorem 2.4 we have

$$(7.3) \quad \beta z(t)[w(t)-1]A(t) + \alpha w(t)[z(t)-1]B(t) = 0 \text{ for all } t.$$

$z(q) = \alpha$, $w(q) = 1$. Hence, for t near q ,

$$(7.4) \quad \begin{aligned} A(t) &= P(z(t), 0) = \frac{r_{-1}}{z(t)-z(q)} + r_0 + \cdots, \\ B(t) &= P(0, w(t)) = \frac{\alpha-1}{\alpha} + \frac{\partial P}{\partial w}(0, 1)(w(t)-1) + \cdots. \end{aligned}$$

Insert (7.4) into (7.3) and expand in powers of $(t-q)$. Though tedious, the computation of the coefficients is straightforward. Equating the first two to 0 yields (7.2).

Proof of Theorem 1. i) Let

$$(7.5) \quad M(z) = \frac{\partial P}{\partial w}(z, 1) = \sum M_m z^m, \quad |z| \leq 1.$$

Then

$$(7.6) \quad M_m = \sum_n np_{mn} = E(X_2|X_1 = m) \cdot \frac{\alpha - 1}{\alpha^{m+1}}.$$

Differentiating (2.2) with respect to w at $w = 1$,

$$(7.7) \quad M(z) = (\alpha - 1) \frac{2z^2 - (1 + \alpha + \beta)z + \alpha}{(z - 1)(z - \alpha)^2} - \frac{\beta z P(z, 0)}{(z - 1)(z - \alpha)} - \frac{\alpha}{z - \alpha} \left[\frac{\alpha - 1}{\alpha} + \frac{\partial P}{\partial w}(0, 1) \right].$$

Expand the right side of (7.7) into partial fractions, and use the lemma. Then

$$(7.8) \quad M(z) = \frac{\alpha(\alpha - 1)}{(\alpha - \beta)(z - \alpha)} + g(z), \quad |z| \leq 1$$

where $g(z)$ is analytic for $|z| < a_3$. Since $\alpha < a_3$, we obtain (1.1) by equating coefficients in (7.8).

ii) Let

$$(7.9) \quad m(w) = \frac{\partial P}{\partial z}(1, w) = \sum m_n w^n, \quad |w| \leq 1.$$

Then

$$(7.10) \quad m_n = \sum_m mp_{mn} = E(X_1|X_2 = n) \cdot \frac{\beta - 1}{\beta^{n+1}}.$$

Differentiating (2.2) with respect to z at $z = 1$,

$$(7.11) \quad m(w) = \frac{C}{w - \beta} + \frac{\beta(\beta - \alpha)}{(w - \beta)^2} + h(w), \quad |w| \leq 1$$

where C is a constant depending on α, β and $h(w)$ is analytic for $|w| < a'_3$. Since $\beta < a'_3$, we obtain (1.2) by equating coefficients in (7.11).

iii) Proceeding as in i) and using (2.10), we get

$$(7.12) \quad M(z) = \frac{\alpha(\alpha - 1)^{1/2}}{(\alpha - z)^{1/2}} + \frac{C_1}{1 - z} \left(\frac{1}{(\alpha - 1)^{1/2}} - \frac{1}{(\alpha - z)^{1/2}} \right) + \frac{C_2}{\alpha - z}, \quad |z| \leq 1.$$

Let $\sum a_m z^m, \sum b_m z^m, \sum c_m z^m$ be the respective power series of

$$(\alpha - z)^{-1/2}, (\alpha - z)^{-3/2}, \frac{1}{1 - z} [(\alpha - 1)^{-1/2} - (\alpha - z)^{-1/2}].$$

Then

$$(7.13) \quad a_m = (-1)^m \binom{-1/2}{m} \frac{1}{\alpha^{m+1/2}}, \quad b_m = 2(m + 1)a_{m+1}, \quad c_m = \sum_{k=m+1}^{\infty} a_k.$$

By Stirling's formula,

$$(7.14) \quad a_m \sim \frac{1}{\sqrt{\pi m} \alpha^{m+1/2}}, \quad b_m \sim \frac{2\sqrt{m}}{\sqrt{\pi} \alpha^{m+3/2}}, \quad c_m = O\left(\frac{1}{\sqrt{m} \alpha^m}\right).$$

Equating coefficients in (7.12) and using (7.14), we get (2.3).

8. Proof of Theorem 2. i) Follows from (6.6).

ii) For $0 < \phi < \infty$, $\sqrt{\alpha/a'_3} < z < a_3$, let

$$v(z, \phi) = (z^\phi w)^{-1}, \quad z(\phi) = z_{1/\phi}, \quad V(\phi) = v(z(\phi), \phi).$$

(6.8) can be restated as follows. If $m/n = \phi$, then

$$(8.1) \quad p_{mn} \sim \frac{C_{1/\phi} V^n(\phi)}{\phi^{1/2} n^{1/2}} \text{ as } n \rightarrow \infty, \quad \text{uniformly for } c \leq \phi \leq d,$$

$$\text{whenever } 0 < c \leq d < \frac{\beta - 1}{\beta - \alpha}.$$

Let $\phi_0 = 1/\theta_0 = (\beta - \alpha)/(\beta - 1)$. We compute C_{θ_0} and obtain the 2nd order Taylor expansion of $V(\phi)$ at ϕ_0 . Since $z(q - 4m) = 1$, $w(q - 4m) = \beta$, we conclude from (4.1) that $q - 4m = t_{\theta_0}$. Hence

$$(8.2) \quad z_{\theta_0} = 1, \quad w_{\theta_0} = \beta, \quad \phi_-(z_{\theta_0}) = 0.$$

A straightforward but tedious computation gives

$$(8.3) \quad \frac{\partial^2 u}{\partial z^2}(1, \theta_0) = \frac{\theta_0 \sigma^2}{\beta^\theta},$$

where σ^2 is defined by (1.6). Hence by (6.9),

$$(8.4) \quad \lim_{\theta \rightarrow \theta_0} C_\theta = C_{\theta_0} = \frac{1}{\sigma \sqrt{2\pi\theta_0}} \cdot \frac{\beta - 1}{\beta}.$$

$z(\phi)$ satisfies $(\partial v / \partial z)(z(\phi), \phi) = 0$. Differentiation rules then give

$$(8.5) \quad V(\phi) = \frac{1}{\beta} \left[1 - \frac{(\phi - \phi_0)^2}{2\sigma^2} \right] + O(|\phi - \phi_0|^3), \quad \phi \text{ near } \phi_0.$$

Let $t = (m - \phi_0 n) / \sqrt{n}$, $\Delta t = 1 / \sqrt{n}$. We conclude from (8.1), (8.4), (8.5) that, for $a < b$,

$$(8.6) \quad \begin{aligned} \lim_{n \rightarrow \infty} P[\phi_0 n + a\sqrt{n} < X_1 < \phi_0 n + b\sqrt{n} | X_2 = n] &= \lim_{n \rightarrow \infty} \sum_{a < t < b} p_{mn} / \frac{\beta - 1}{\beta^{n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{1}{\sigma \sqrt{2\pi}} \sum_{a < t < b} \left[1 - \frac{t^2}{2\sigma^2 n} + O\left(\frac{1}{n^{3/2}}\right) \right]^n \Delta t \\ &= \lim_{n \rightarrow \infty} \frac{1}{\sigma \sqrt{2\pi}} \sum_{a < t < b} e^{-t^2/2\sigma^2} (\Delta t + O(\{\Delta t\}^2)) = \frac{1}{\sigma \sqrt{2\pi}} \int_a^b e^{-t^2/2\sigma^2} dt. \end{aligned}$$

iii) For $\alpha = \beta$, $z(p - 4m) = \alpha$ and $w(p - 4m) = 1$. Hence by (4.1), $p - 4m = t_0$. By (6.11),

$$(8.7) \quad \lim_{\theta \rightarrow 0} D_\theta = D_0 = \frac{1}{\sqrt{\pi}} \left(\frac{\alpha - 1}{\alpha} \right)^{3/2}.$$

u_θ satisfies $(\partial u / \partial w)(w_\theta, \theta) = 0$. Differentiation rules then give

$$(8.8) \quad u_\theta = \frac{1}{\alpha} \left[1 - \frac{\theta^2}{2\sigma^2} \right] + O(|\theta|^3), \quad \theta \text{ near } 0$$

where σ^2 is defined by (1.9).

Let $t = n/\sqrt{m}$, $\Delta t = 1/\sqrt{m}$, $x > 0$. We conclude from (6.10), (8.7), (8.8) that

$$\begin{aligned}
 (8.9) \quad & \lim_{m \rightarrow \infty} P[X_2 < x\sqrt{m} | X_1 = m] \\
 &= \lim_{m \rightarrow \infty} \sum_{t < x} p_{mn} \frac{\alpha - 1}{\alpha^{m+1}} \\
 &= \lim_{m \rightarrow \infty} \frac{1}{\sigma} \sqrt{\frac{2}{\pi}} \sum_{0 \leq t < x} \left(1 - \frac{t^2}{2\sigma^2 m} + O\left(\frac{1}{m^{3/2}}\right) \right)^m \Delta t \\
 &= \lim_{m \rightarrow \infty} \frac{1}{\sigma} \sqrt{\frac{2}{\pi}} \sum_{0 \leq t < x} e^{-t^2/2\sigma^2} (\Delta t + O(\Delta t^2)) = \frac{1}{\sigma} \sqrt{\frac{2}{\pi}} \int_0^x e^{-t^2/2\sigma^2} dt.
 \end{aligned}$$

Appendix—The mapping properties of $u(t, \theta)$. Let γ be the positively oriented boundary of R_1 . γ consists of the four sides γ_j , $1 \leq j \leq 4$; γ_j is labeled in Fig. 6i) by j . We describe $u(\gamma, \theta)$, the image of γ under $u(t, \theta)$ for fixed θ , $0 < \theta < 1$. An application of the argument principle then gives Theorem 5.1.

LEMMA 1. Let $\xi(t) = 1/(z(t)w(t))$. $\xi(t) = z(t+2m)/\alpha$.

Proof. From [2, Thm. 3.3], we conclude that $\xi(t)$, $z(t+2m)$ have identical zeros and poles. Hence

$$1) \quad z(t+2m) = c\xi(t), \quad c \text{ a constant} \neq 0.$$

c is evaluated as follows. $z(t)$ has a simple zero at $-m$. Hence

$$2) \quad z(m+\tau) = z(-m-\tau) \sim -z(-m+\tau) \text{ as } \tau \rightarrow 0.$$

By [2, formula (3.9)]

$$3) \quad w(-m+\tau) \sim -\alpha/(z^2(-m+\tau)) \text{ as } \tau \rightarrow 0.$$

Let $t = -m + \tau$ in 1) and use 2), 3).

$$4) \quad c = \lim_{\tau \rightarrow 0} z(-m+\tau)z(m+\tau)w(-m+\tau) = \alpha.$$

Let $w_1(t) = w(-\omega + t)$, $w_3(t) = w(-2m + t)$, $0 \leq t \leq \tilde{\omega}$. $w_1(t)$, $w_3(t)$ are the w -images of the lower and upper sides of R_1 . (See Fig. 6ii) where $w(\gamma_j)$ is labeled by j .) From the mapping properties of w we conclude that: $w_1(0) = 0$, $w_1(\tilde{\omega}) > 0$, $w_3(0) < 0$, $w_3(\tilde{\omega}) > 0$, and, for all other t , $w_1(t)$ lies in the interior of the upper half plane. Hence $\theta_1(t) = \arg w_1(t)$, $\theta_3(t) = \arg w_3(t)$ can be chosen to be continuous functions of t with range in $[0, \pi]$.

LEMMA 2. i) $\theta_1(t)$ is a strictly decreasing function of t , with $\theta_1(0) = (\pi/2)$, $\theta_1(\tilde{\omega}) = 0$.
ii) $\theta_3(t)$ is a strictly decreasing function of t , with $\theta_3(t) = \pi$, $\theta_3(\tilde{\omega}) = 0$.

From the monotonicity of $\theta_1(t)$, we conclude that $w(\gamma_1)$ lies in the first quadrant of the w -plane.

Proof. i) The statements concerning the values of w_1 , w_3 at 0, $\tilde{\omega}$ follow from the mapping properties of $w(t)$. We show that $\theta_1(t)$ is a strictly decreasing function of t . $z_1(t) = z(-\omega + t)$ is a strictly decreasing function of t mapping $[0, \tilde{\omega}]$ onto $[\infty, a_3]$. Since the discriminant $D_1(z) < 0$ on (a_3, ∞) , we conclude that $Q(z_1(t), w) = 0$ has the two roots $w_1(t)$, $\bar{w}_1(t)$ satisfying

$$5) \quad w_1 \bar{w}_1 = |w_1|^2 = \frac{\beta}{z_1}, \quad w_1 + \bar{w}_1 = 2|w_1| \cos \theta_1 = \frac{(1 + \alpha + \beta)z_1 - \alpha}{z_1^2}.$$

Hence, for $a_3 \leq z_1 < \infty$,

$$6) \quad \cos \theta_1 = \frac{(1 + \alpha + \beta)z_1 - \alpha}{2\beta^{1/2}z_1^{3/2}},$$

$$7) \quad \frac{d}{dz_1}(\cos \theta_1) = \frac{1 + \alpha + \beta}{4\beta^{1/2}z_1^{5/2}} \left(\frac{3\alpha}{1 + \alpha + \beta} - z_1 \right),$$

the square roots in 6), 7) being > 0 .

A computation gives $D_1(3\alpha/(1 + \alpha + \beta)) > 0$, so that $3\alpha/(1 + \alpha + \beta) < a_3$. It follows from (7) that $\cos \theta_1$ is a strictly decreasing function of z_1 , which is the same as saying that θ_1 is a strictly decreasing function of t on $(0, \tilde{\omega}]$. From 6), $\lim_{z_1 \rightarrow \infty} \cos \theta_1 = 0$ which is the same as $\theta_1(0) = \pi/2$. $\theta_1(\tilde{\omega}) = 0$ follows from $w_1(\tilde{\omega}) > 0$.

ii) Let $\tilde{Q}(\xi, w) = \xi^2 w Q(1/\xi w, w)$, and let $\tilde{D}_1(\xi)$ be the ξ -discriminant of $\tilde{Q}$. Then $\tilde{D}_1(\xi) = D_1(\alpha\xi)/\alpha^2$. We have $\tilde{Q}(\xi(t), w(t)) = 0$. By Lemma 1, $\xi_3(t) = \xi(-2m + t) = z(t)/\alpha$ is a strictly increasing function of t mapping $[0, \tilde{\omega}]$ onto $[a_1/\alpha, a_2/\alpha]$. $\tilde{D}_1(\xi) < 0$ on $(a_1/\alpha, a_2/\alpha)$, so that $\tilde{Q}(\xi_3(t), w) = 0$ has the two roots $w_3(t), \bar{w}_3(t)$. A computation similar to that of i) gives

$$8) \quad \frac{d}{d\xi_3}(\cos \theta_3) = \frac{1 + \alpha + \beta}{4\alpha^{1/2}\beta^{1/2}\xi_3^{5/2}} \left[\frac{3}{1 + \alpha + \beta} - \xi_3 \right], \quad \frac{a_1}{\alpha} \leq \xi_3 \leq \frac{a_2}{\alpha},$$

the square roots in 8) being > 0 . A computation gives $D_1(\alpha/(1 + \alpha + \beta)) < 0$, so that $a_1 < \alpha/(1 + \alpha + \beta) < 3\alpha/(1 + \alpha + \beta)$. As $D_1(3\alpha/(1 + \alpha + \beta)) < 0$, we conclude that $a_2 < 3\alpha/(1 + \alpha + \beta)$. It follows from 8) that $\cos \theta_3$ is a strictly increasing function of ξ_3 , which is the same as saying that θ_3 is a strictly decreasing function of t . $\theta_3(0) = \pi$, $\theta_3(\tilde{\omega}) = 0$ follow from $w_3(0) < 0$, $w_3(\tilde{\omega}) > 0$.

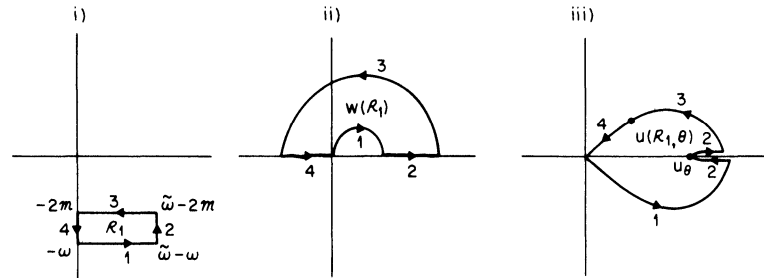


FIG. 6

Proof of Theorem 5.1. We describe $u(\gamma_j, \theta)$ for $1 \leq j \leq 4$. (See Fig. 6iii) where $u(\gamma_j, \theta)$ is labeled by j .)

- γ_1 : Since $z > 0$ along γ_1 and $u = (zw^\theta)^{-1}$, we have $\arg u = -\theta \arg w$. We conclude from Lemma 2 that $u(\gamma_1, \theta)$ is a simple curve lying, except for initial and end points, in the interior of the fourth quadrant of the u -plane. The initial point is 0 and the end point is positive.
- γ_2 : By Theorem 4.2, $u(\gamma_2, \theta)$ consists of two straight line segments on the positive real axis, both having u_θ for left end point.
- γ_3 : By Lemma 1, $\xi(t) > 0$ along γ_3 . Since $u = \xi w^{1-\theta}$, $\arg u = (1 - \theta) \arg w$. We conclude from Lemma 2 that $u(\gamma_3, \theta)$ is a simple curve lying, except for initial point, in the interior of the upper half u -plane.

γ_4 : Parametrize γ_4 by $t = -2m - m\tau$, $0 \leq \tau \leq 1$. Let $u = r e^{i\psi}$, $r = |u|$. Use the mapping properties of z , w and Theorem 4.1 to conclude from (4.3) that

$$9) \quad \frac{du}{dt} \bigg/ u = \frac{dr}{dt} \bigg/ r + i \frac{d\psi}{d\tau} < 0, \quad 0 \leq \tau < 1.$$

9) implies that $dr/d\tau < 0$ and ψ is constant along γ_4 . It follows that $u(\gamma_4, \theta)$ is a straight line segment going from $u(-2m, \theta)$ to 0.

We conclude from the descriptions of $u(\gamma_j, \theta)$, $1 \leq j \leq 4$, that $u(\gamma, \theta)$ is a closed curve, which is nonintersecting except for the straight line segment $u(\gamma_2, \theta)$ which folds over on itself. By the argument principle $u(t, \theta)$ maps the interior of R_1 conformally onto the open region D bounded by $u(\gamma, \theta)$. Denote the inverse map by $s(u, \theta)$. Since $(0, u_\theta) \subset D$, $t(u, \theta)$ is analytic for $0 < u < u_\theta$. A slight modification of the usual inverse function theorem shows that $t(u, \theta)$ can be extended as a continuous function to $0 \leq u \leq u_\theta$ by letting $t(0, \theta) = 0$, $t(u_\theta, \theta) = t_\theta$.

REFERENCES

- [1] N. G. DE BRUIJN, *Asymptotic Methods in Analysis*, third edition, North-Holland, Amsterdam, 1949.
- [2] L. FLATTO AND S. HAHN, *Two parallel queues created by arrivals with two demands I*, this Journal, 44 (1984), pp. 1041–1054; *Erratum*, 45 (1985), p. 168.
- [3] *Handbook of Mathematical Functions*, National Bureau of Standards, Applied Math. Ser. 55, Tenth Printing, 1972.
- [4] V. A. MALYSHEV, *An analytic method in the theory of two-dimensional positive random walks*, Sibirskii Math. Zh., 13 (1972), pp. 1314–1329.
- [5] ———, *Asymptotic behavior of stationary probabilities for two-dimensional positive random walks*, Sibirskii Math. Zh., 14 (1973), pp. 109–117.
- [6] S. SAKS AND A. SIGMUND, *Analytic Functions*, 2nd edition, Warsaw, 1965.